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Question No : 41 of 52 Marks: 2 (Budgeted Time 4 Min)

Use Wallis sine formula to evaluate $\int_0^{\frac{\pi}{2}} \sin^5 x \, dx$

Answer (Please click here to Add Answer)

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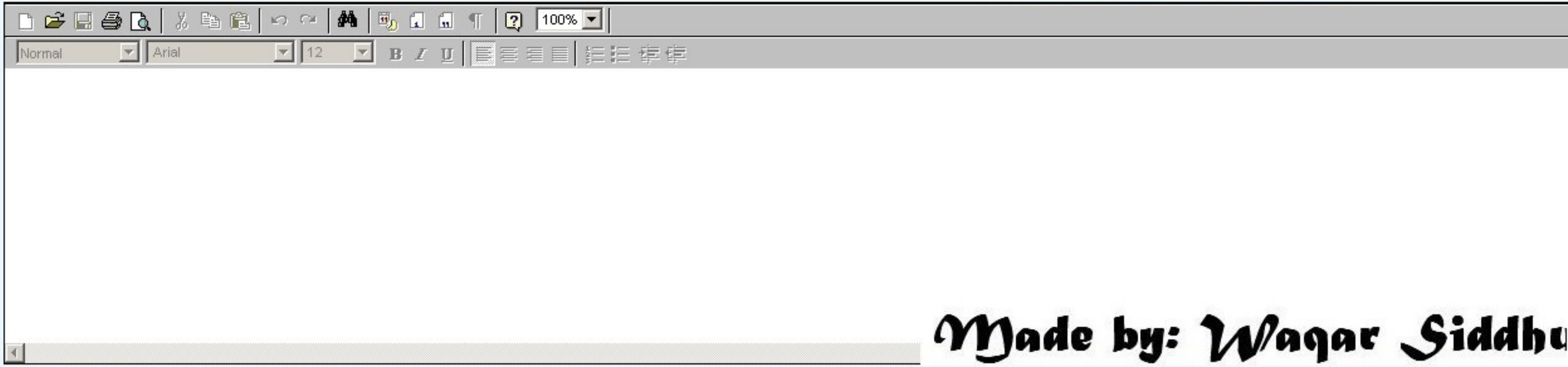
Question No : 42 of 52 Marks: 2 (Budgeted Time 4 Min)

Prove whether the following function is even, odd or neither.

$f(x) = x^3 + x^2$

Answer (Please click here to Add Answer)

VuAnswers.com



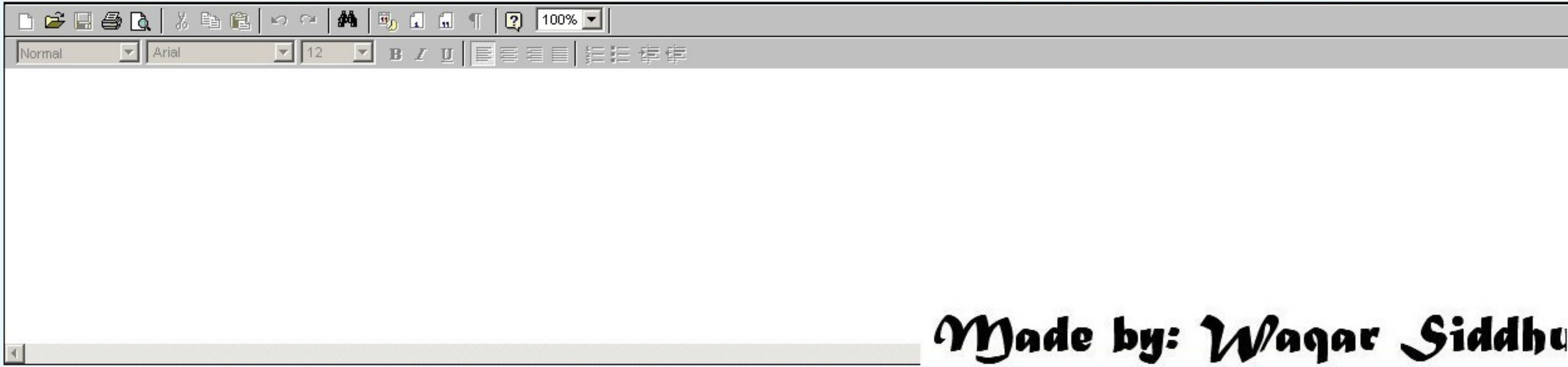
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Question No : 43 of 52 Marks: 2 (Budgeted Time 4 Min)

Consider the point $\left(\frac{\sqrt{3}}{2}, \sqrt{3}, \frac{1}{2}\right)$ in rectangular coordinate system. Find the value of "ρ" in spherical coordinates.

Answer (Please click here to Add Answer)

VuAnswers.com



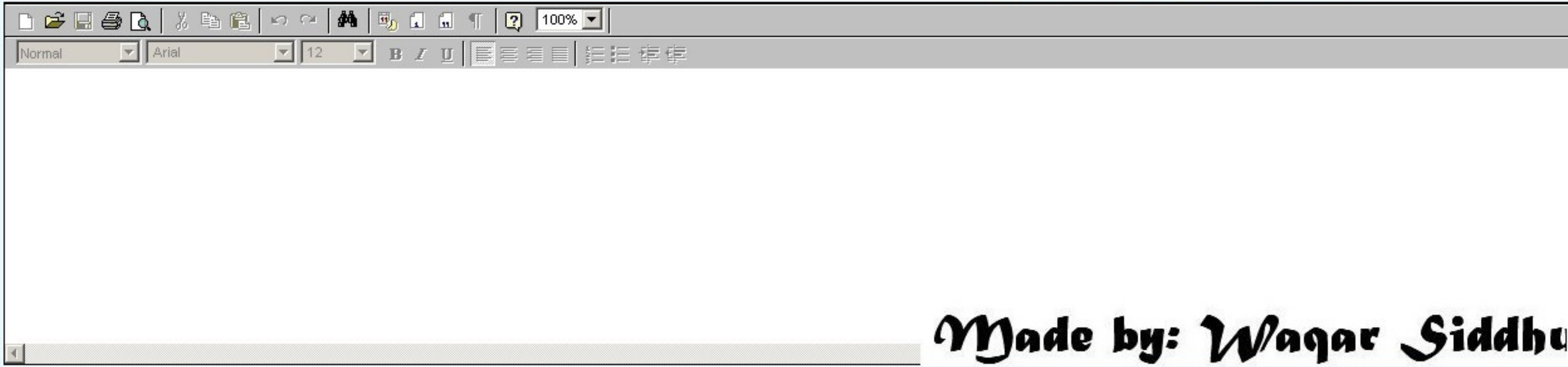
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Question No : 44 of 52 Marks: 2 (Budgeted Time 4 Min)

Write down the expression for the arc- length of the curve represented by the vector valued function $\vec{r}(t) = \cos t^2 \hat{i} + \sin t^2 \hat{j}$ where $0 \le t \le \frac{\pi}{2}$. (Do not evaluate the expression).

Answer (Please click here to Add Answer)

VuAnswers.com



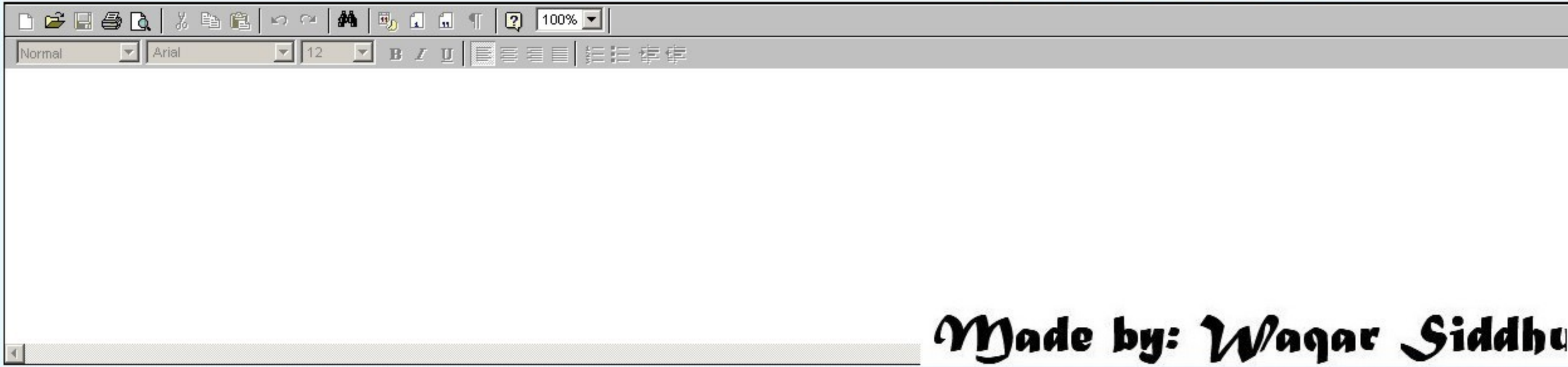
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Question No : 45 of 52 Marks: 3 (Budgeted Time 6 Min)

Use Wallis cosine formula to evaluate $\int_0^{\frac{\pi}{2}} \left(\cos^3 x + \cos^4 x\right) dx$

Answer (Please click here to Add Answer)

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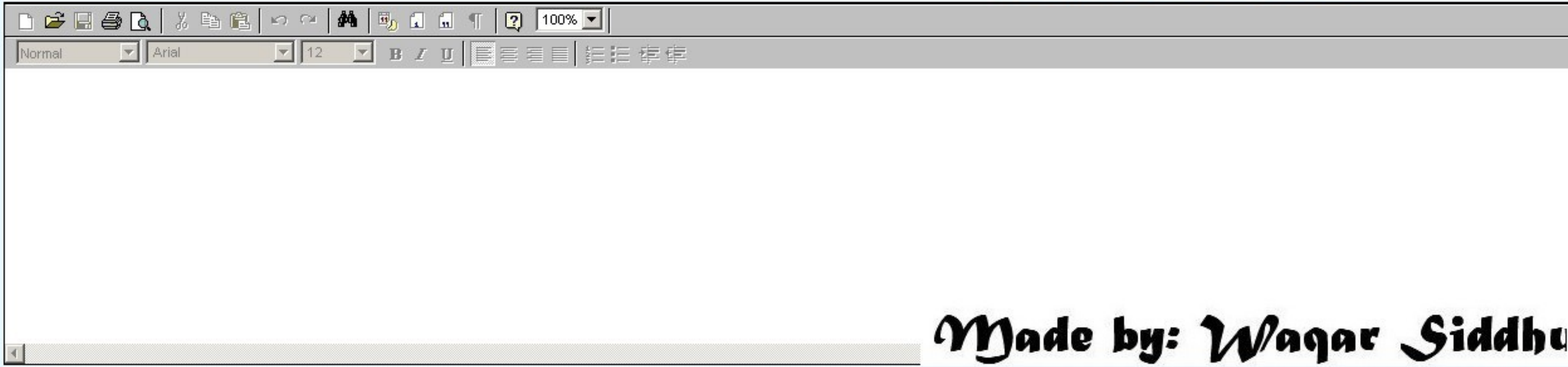
Question No : 46 of 52 Marks: 3 (Budgeted Time 6 Min)

Prove whether the following function is even, odd or neither.

$f(x) = \sin^2 x \cos 3x$

Answer (Please click here to Add Answer)

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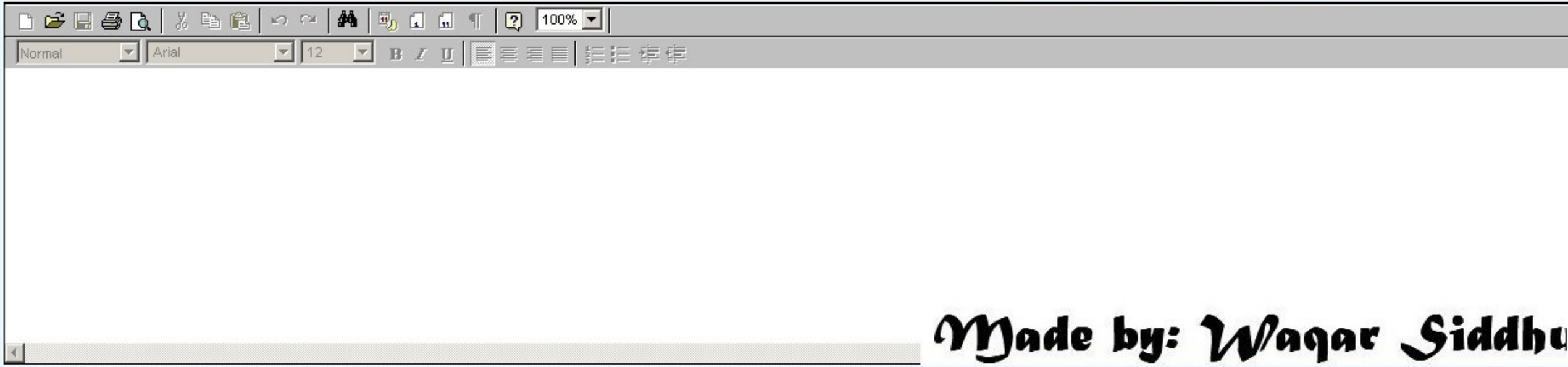
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Question No : 47 of 52 Marks: 3 (Budgeted Time 6 Min)

Find the region where the function $f(x, y) = \sqrt{9 - x^2 - y^2}$ is continuous.

Answer (Please click here to Add Answer)

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Question No : 48 of 52 Marks: 3 (Budgeted Time 6 Min)

What is the arc-length of the curve $\vec{r}(t) = (4+3t)\,\hat{i} + (2-2t)\,\hat{j}$ when $3 \le t \le 4$?

Answer (Please click here to Add Answer)

VuAnswers.com

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Question No : 49 of 52 Marks: 5 (Budgeted Time 10 Min)

Determine the Fourier co-efficient b_n of the periodic function defined below:

$f(x) = 2x + 1 \qquad 0 < x < 2$

Answer (Please click here to Add Answer)

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Question No : 50 of 52 Marks: 5 (Budgeted Time 10 Min)

Determine whether the following vector field $\vec{F}$ is conservative or not.

$\vec{F}(x, y, z) = (3x + y)\hat{i} + xy^2z\,\hat{j} + xz^2\,\hat{k}$

Answer (Please click here to Add Answer)

VuAnswers.com

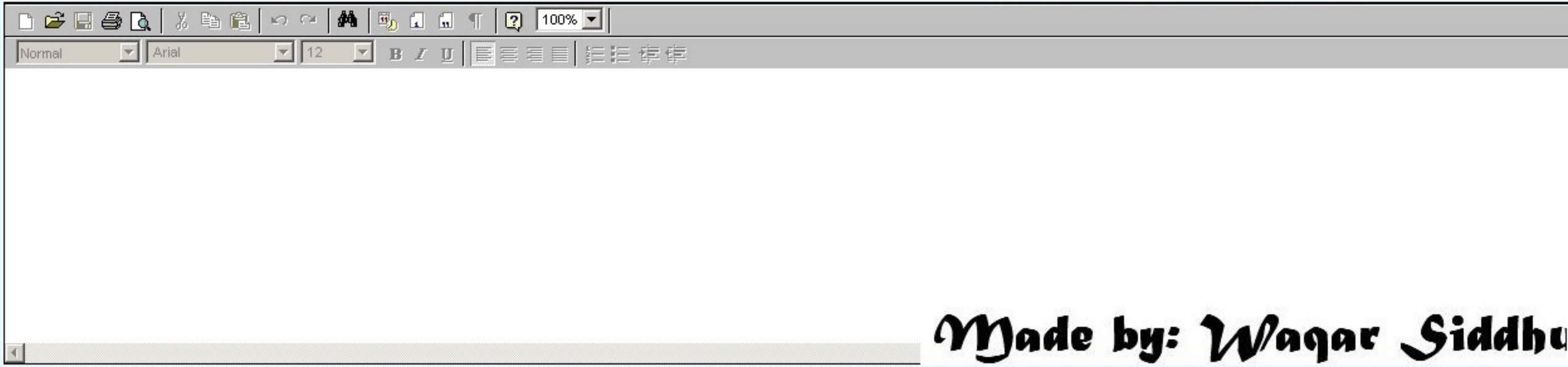
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Question No : 51 of 52 Marks: 5 (Budgeted Time 10 Min)

Using second partial derivative test show that the function $f(x, y) = xy(24 - x - y)$ is maximum at $x = 8, y = 8$.

Answer (Please click here to Add Answer)

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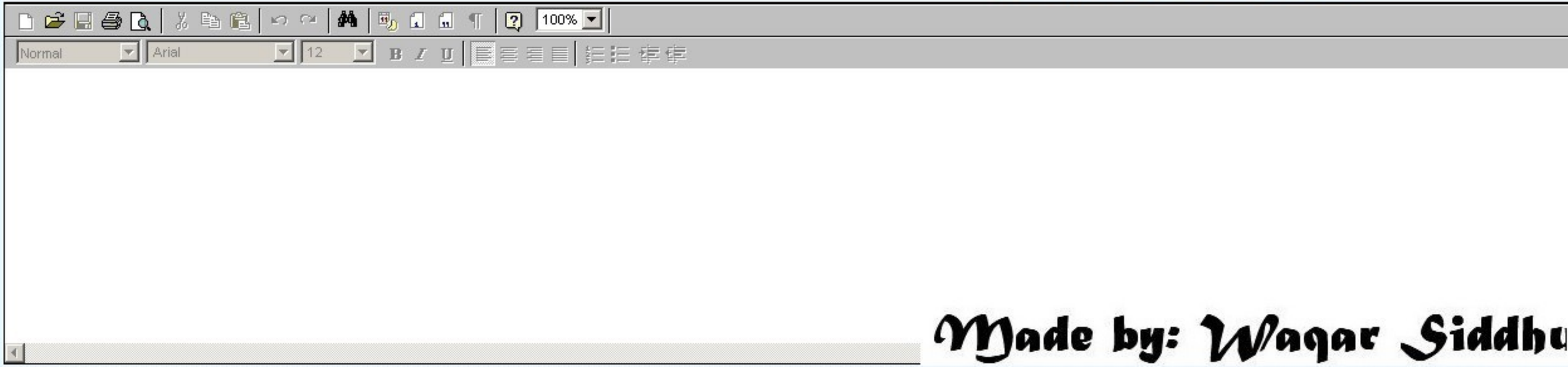
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Question No : 52 of 52 Marks: 5 (Budgeted Time 10 Min)

Let $\vec{r}(t) = t^2\,\hat{i} + t\,\hat{j} + (t^2 - 5)\,\hat{k}$. Find t, such that $\vec{r}(t)$ and $\vec{r}'(t)$ are perpendicular to each other.

Answer (Please click here to Add Answer)

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Question No : 41 of 52 Marks: 2 (Budgeted Time 4 Min)

Determine whether the following differential is exact or not.

$dz = 4x^3y^3\ dx + 3x^4y^2\ dy$

Answer (Please click here to Add Answer)

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Question No : 42 of 52 Marks: 2 (Budgeted Time 4 Min)

Evaluate

$$\int_{-\pi}^{\pi} \cos nx \, dx$$

where n is an integer other than zero.

Answer (Please click here to Add Answer)

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Made by: Waqar Siddhu

Question No : 43 of 52 Marks: 2 (Budgeted Time 4 Min)

Prove whether the following function is even, odd or neither.

$f(x) = x^2 - 4\sin x$

Answer (Please click here to Add Answer)

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Question No : 44 of 52 Marks: 2 (Budgeted Time 4 Min)

Given $\vec{a} \times \vec{b} = 3xi + 2yj + xk$ and $\vec{c} = 7xi + 4yk$. Find scalar triple product of these vectors.

VuAnswers.com

Answer (Please click here to Add Answer)

Made by: Waqar Siddhu

Question No : 45 of 52 Marks: 3 (Budgeted Time 6 Min)

What is the arc-length of the curve $\vec{r}(t) = (4+3t)\,\hat{i} + (2-2t)\,\hat{j} + (5+t)\,\hat{k}$ when $3 \le t \le 4$?

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Answer (Please click here to Add Answer)

Made by: Waqar Siddhu

Question No : 46 of 52 Marks: 3 (Budgeted Time 6 Min)

Find $div\ \vec{F}$, if $\vec{F} = (3x + y)\hat{i} + xy^2z\,\hat{j} + (xz^2)\hat{k}$

Answer (Please click here to Add Answer)

VuAnswers.com

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Question No : 47 of 52 Marks: 3 (Budgeted Time 6 Min)

Determine the Fourier co-efficient a_0 of the periodic function defined below:

$f(x) = 2x + 1 \qquad 0 < x < 2$

Answer (Please click here to Add Answer)

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Question No : 48 of 52 Marks: 3 (Budgeted Time 6 Min)

A line, in three dimensional space, passes through the point $(3, -4, 2)$ and parallel to the vector $\vec{n} = 4\hat{i} + 3\hat{j} + 6\hat{k}$. Write down the equation of this line in parametric and symmetric form.

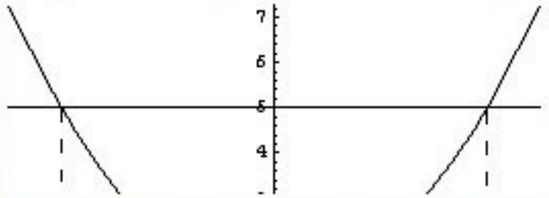
Answer (Please click here to Add Answer)

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Question No : 49 of 52 Marks: 5 (Budgeted Time 10 Min)

Using definite integral, find area of the region bounded by the curves of $y = x^2 + 1$ *and* $y = 5$



Answer (Please click here to Add Answer)

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Question No : 50 of 52 Marks: 5 (Budgeted Time 10 Min)

Show that Laplace transform of the function

$f(t) = 1$ is

$\frac{1}{s}$ where s is a constant for the integration and $s > 0$.

Answer (Please click here to Add Answer)

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Question No : 51 of 52 Marks: 5 (Budgeted Time 10 Min)

Determine whether the following vector field $\vec{F}$ is conservative or not.

$\vec{F}(x, y, z) = (4x - z)\hat{i} + (3y + z)\hat{j} + (y - x)\hat{k}$

Answer (Please click here to Add Answer)

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Question No : 52 of 52 Marks: 5 (Budgeted Time 10 Min)

Consider the point $(-5, 5, 6)$ in rectangular coordinate system. Convert it into Spherical coordinates.

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Answer (Please click here to Add Answer)

Made by: Waqar Siddhu

Question No : 41 of 52 Marks: 2 (Budgeted Time 4 Min)

Use Wallis cosine formula to evaluate $\int_0^{\frac{\pi}{2}} \cos^6 x \, dx$

Answer (Please click here to Add Answer)

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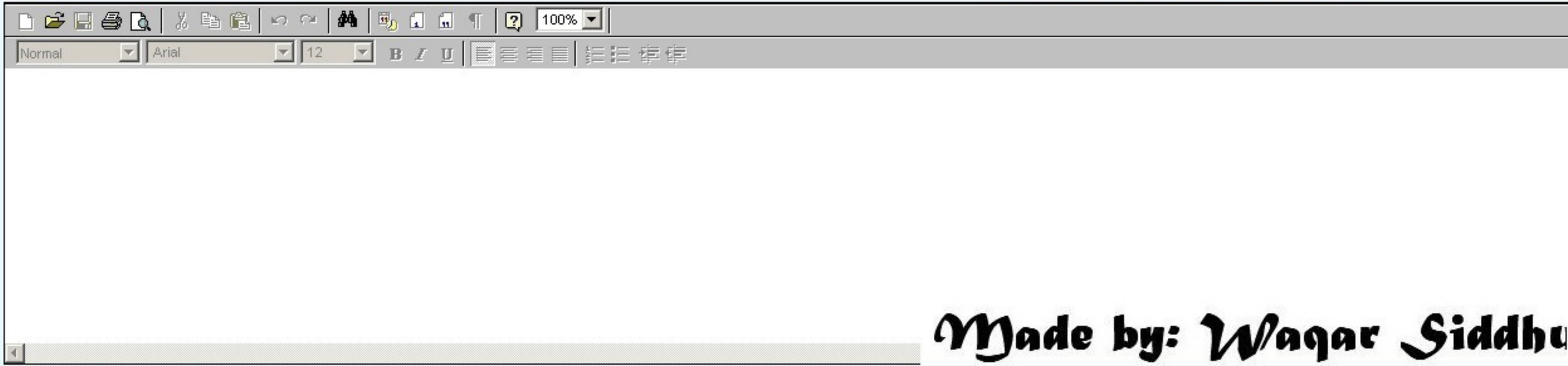
Question No : 42 of 52 Marks: 2 (Budgeted Time 4 Min)

Prove whether the following function is even, odd or neither.

$f(x) = x^3 + x^2$

Answer (Please click here to Add Answer)

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Made by: Waqar Siddhu

Question No : 43 of 52 Marks: 2 (Budgeted Time 4 Min)

Let $f(x, y) = \tan^{-1}\frac{y}{x} - y^2 \tan^{-1}\frac{x}{y}$. Is the function defined at (1, 1)? If yes, what is its value and if no, give the reason.

Answer (Please click here to Add Answer)

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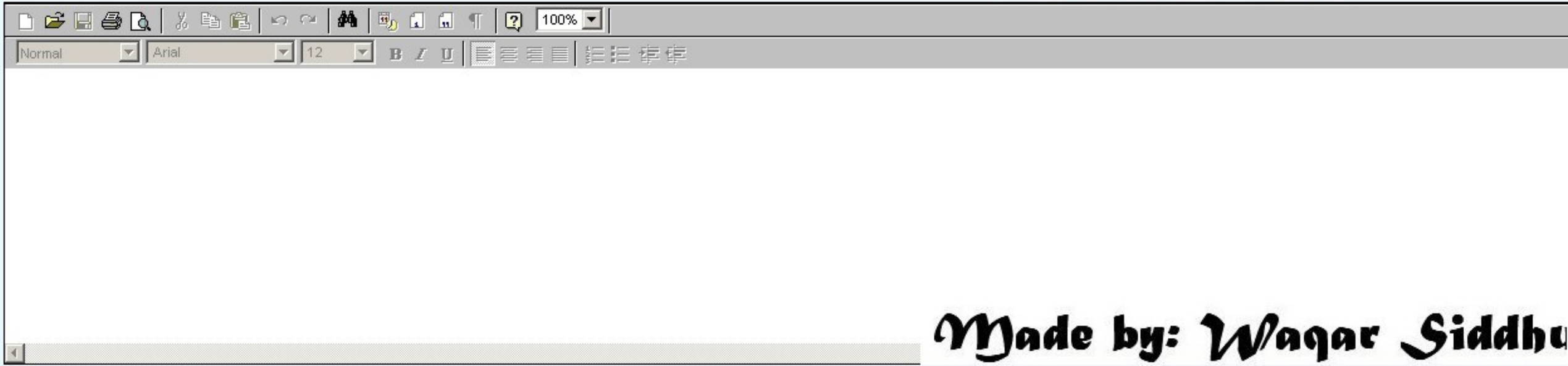
Question No : 44 of 52 Marks: 2 (Budgeted Time 4 Min)

Evaluate the following limit.

$$\lim_{t \to \frac{\pi}{4}} \left[(\cos t)\hat{i} + (\sin t)\hat{j} \right]$$

Answer (Please click here to Add Answer)

VuAnswers.com



Made by: Waqar Siddhu

Question No : 45 of 52 Marks: 3 (Budgeted Time 6 Min)

Determine the fourier co-efficient a_0, of periodic function defined by

$f(x) = x \qquad 0 < x < 1$

Answer (Please click here to Add Answer)

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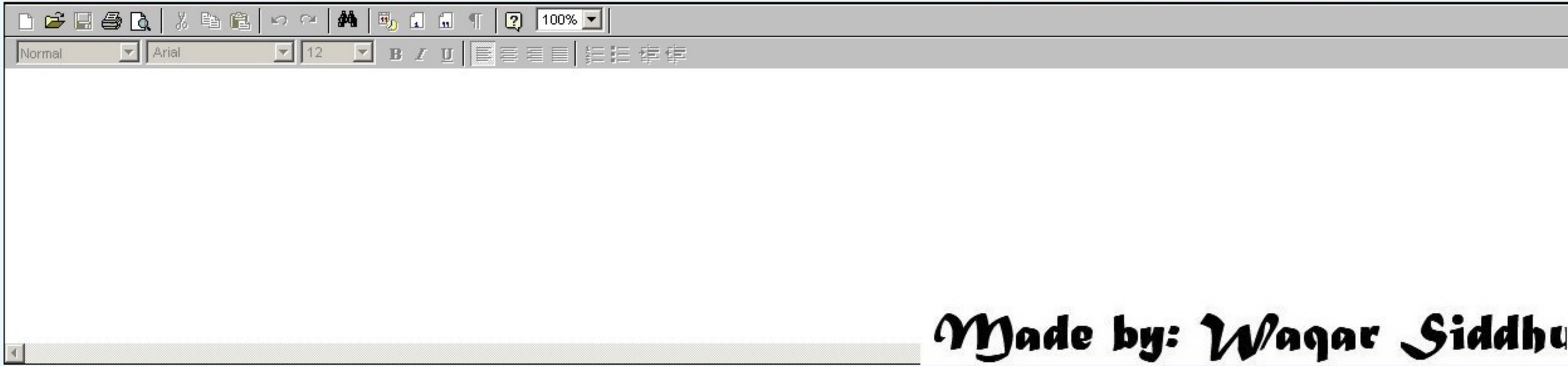
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Question No : 46 of 52 Marks: 3 (Budgeted Time 6 Min)

Use Wallis sine formula to evaluate $\int_0^{\frac{\pi}{2}} \left(\sin^3 x + \sin^4 x\right) dx$

Answer (Please click here to Add Answer)

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Question No : 47 of 52 Marks: 3 (Budgeted Time 6 Min)

If the order of integration for the integral $\int_0^2 \int_{y^2}^4 y \cos x^2 dx dy$ is changed. Find the change in the limits of new integral.

Answer (Please click here to Add Answer)

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Question No : 48 of 52 Marks: 3 (Budgeted Time 6 Min)

What is the arc-length of the curve $\vec{r}(t) = 3\cos t\,\hat{i} + 3\sin t\,\hat{j}$ when $0 \le t \le 2\pi$?

Answer (Please click here to Add Answer)

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Question No : 49 of 52 Marks: 5 (Budgeted Time 10 Min)

Consider a periodic function defined by

$f(x) = 3x \qquad -\pi \le x \le \pi$

(i) Find whether the given function is even or odd?

(ii) Determine Fourier Co-efficients a_0, a_n and b_n

Answer (Please click here to Add Answer)

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Question No : 50 of 52 Marks: 5 (Budgeted Time 10 Min)

Determine whether the following vector field $\vec{F}$ is conservative or not.

$\vec{F}(x, y, z) = (3x + y)\hat{i} + xy^2z\,\hat{j} + xz^2\,\hat{k}$

Answer (Please click here to Add Answer)

VuAnswers.com

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Question No : 51 of 52 Marks: 5 (Budgeted Time 10 Min)

Find Equation of a Tangent plane to the surface $f(x, y, z) = x^2 + 3y + z^3 - 9$ at the point $(2, -1, 2)$

Answer (Please click here to Add Answer)

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Question No : 52 of 52 Marks: 5 (Budgeted Time 10 Min)

Let $\vec{r}(t) = t^2\,\hat{i} + t\,\hat{j} + (t^2 - 5)\,\hat{k}$. Find t, such that $\vec{r}(t)$ and $\vec{r'}(t)$ are perpendicular to each other.

Answer (Please click here to Add Answer)

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Question No : 41 of 52 Marks: 2 (Budgeted Time 4 Min)

Use Wallis sine formula to evaluate $\int_0^{\frac{\pi}{2}} \sin^5 x \, dx$

Answer (Please click here to Add Answer)

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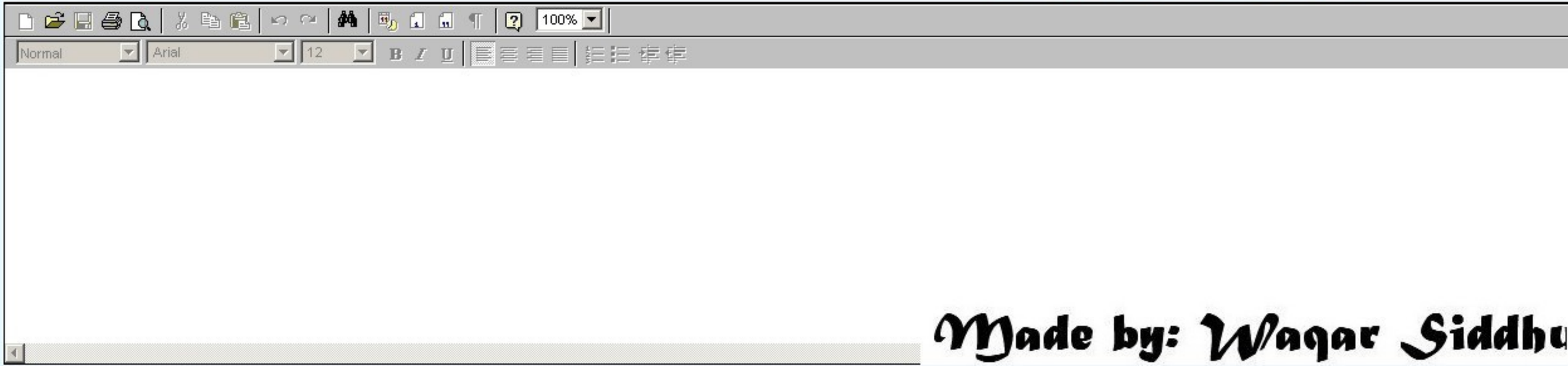
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Question No : 42 of 52 Marks: 2 (Budgeted Time 4 Min)

Find Laplace transform of the function $F(t)$ if $F(t) = \cos 3t$.

Answer (Please click here to Add Answer)

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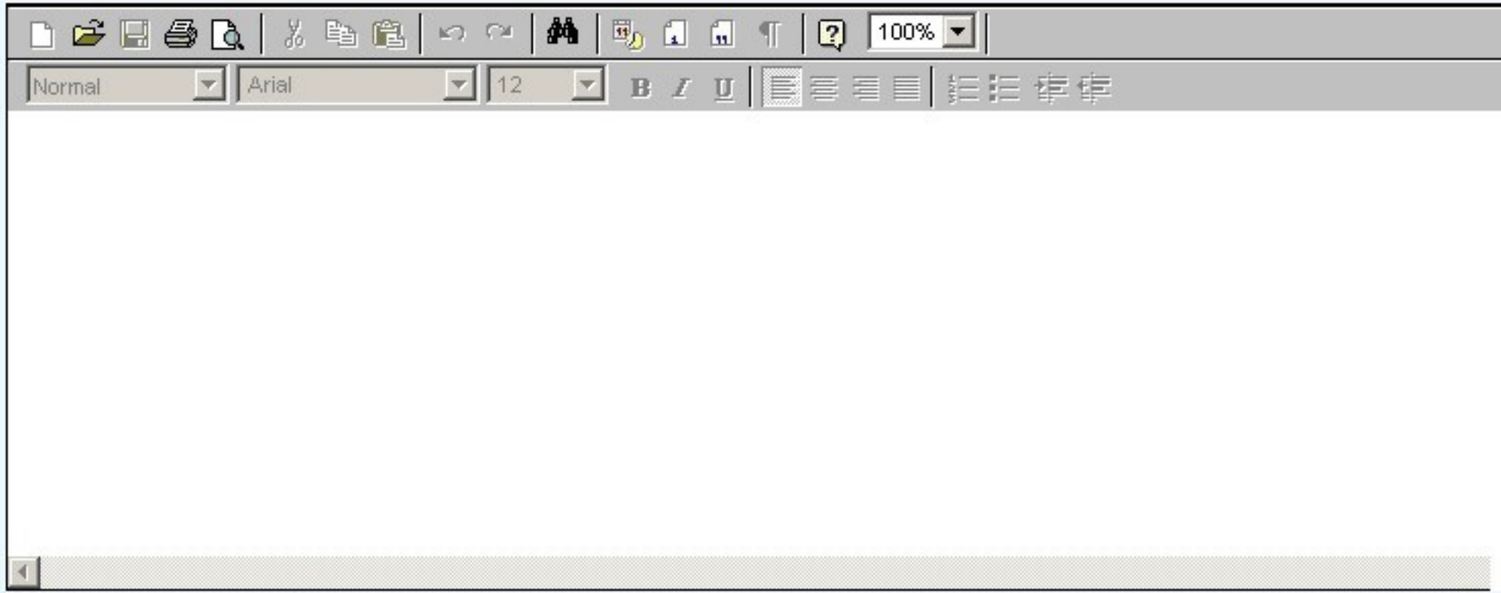
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Question No : 43 of 52 Marks: 2 (Budgeted Time 4 Min)

State the condition when $\iint_R f(x,y)dA = \int_c^d \int_a^b f(x,y)dxdy = \int_a^b \int_c^d f(x,y)dydx$ where R is the region of integration.

Answer (Please click here to Add Answer)

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Question No : 44 of 52 Marks: 2 (Budgeted Time 4 Min)

Find derivative of the following vector-valued function.

$$\vec{r}(t) = e^{t^2}\,\hat{i} + t^2\,\hat{j} + \sec 2t\,\hat{k}$$

Answer (Please click here to Add Answer)

VuAnswers.com



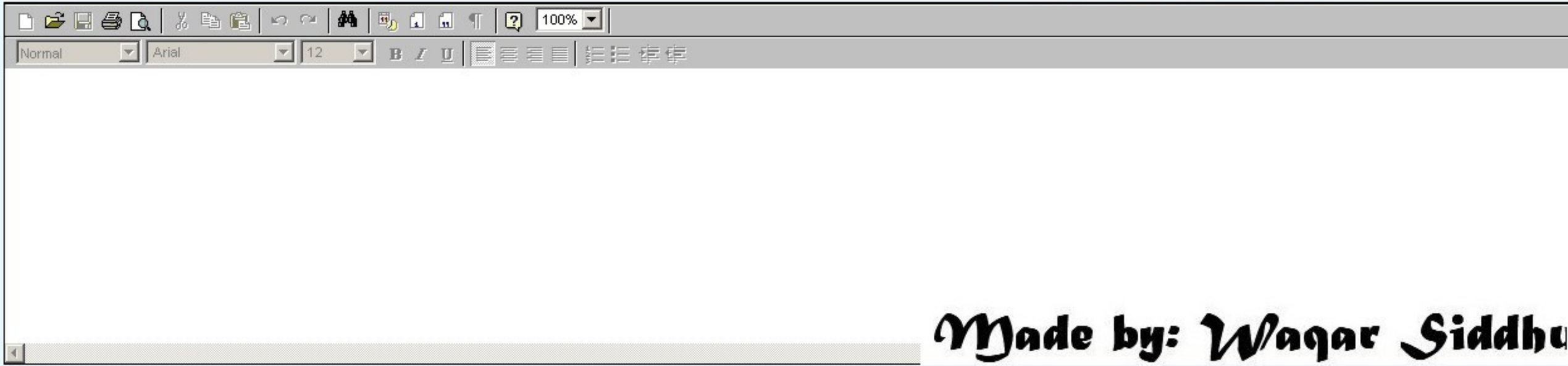
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Question No : 45 of 52 Marks: 3 (Budgeted Time 6 Min)

Use Wallis sine formula to evaluate $\int_0^{\frac{\pi}{2}} \left(\sin^3 x + \sin^4 x\right) dx$

Answer (Please click here to Add Answer)

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Question No : 46 of 52 Marks: 3 (Budgeted Time 6 Min)

Find Laplace transform of the function $F(t)$ if

$F(t) = e^{2t} \sin 3t$

VuAnswers.com

Answer (Please click here to Add Answer)

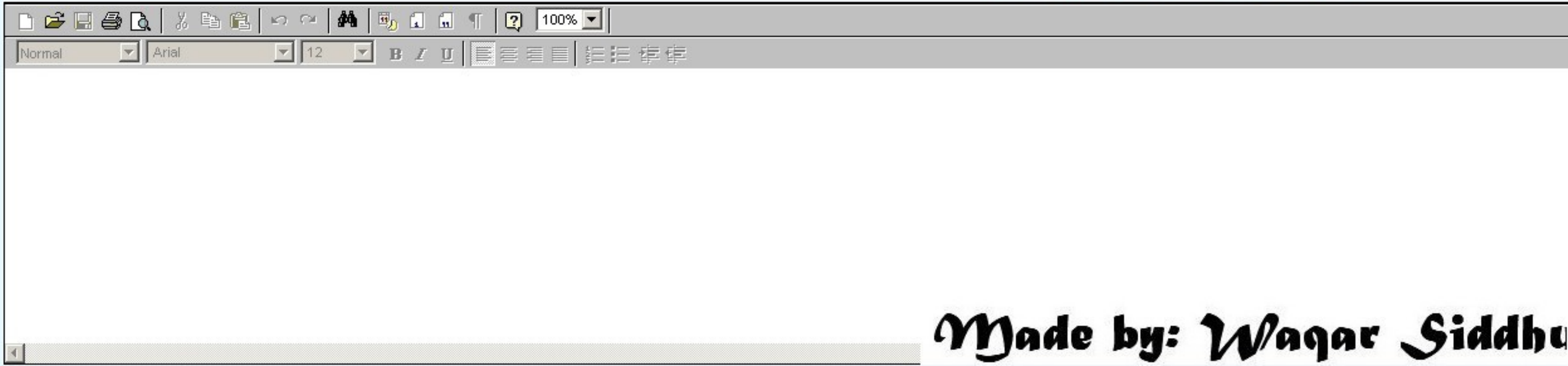
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Question No : 47 of 52 Marks: 3 (Budgeted Time 6 Min)

Find the critical point for the given function $f(x,y) = 6x^2 + xy - 2y^2$ along the line $y = 3x + 1$ at which the absolute extrema of the function can occur.

Answer (Please click here to Add Answer)

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Question No : 48 of 52 Marks: 3 (Budgeted Time 6 Min)

What is the arc-length of the curve $\vec{r}(t) = 3\cos t\,\hat{i} + 3\sin t\,\hat{j}$ when $0 \le t \le 2\pi$?

Answer (Please click here to Add Answer)

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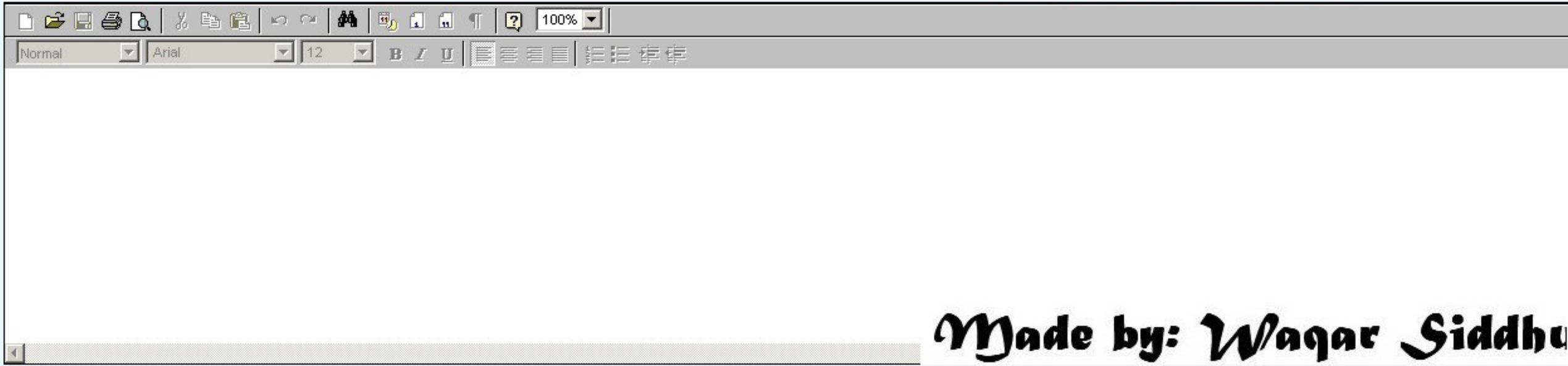
Question No : 49 of 52 Marks: 5 (Budgeted Time 10 Min)

Determine the Fourier co-efficient b_n of the periodic function defined below:

$f(x) = 2x + 1 \qquad 0 < x < 2$

Answer (Please click here to Add Answer)

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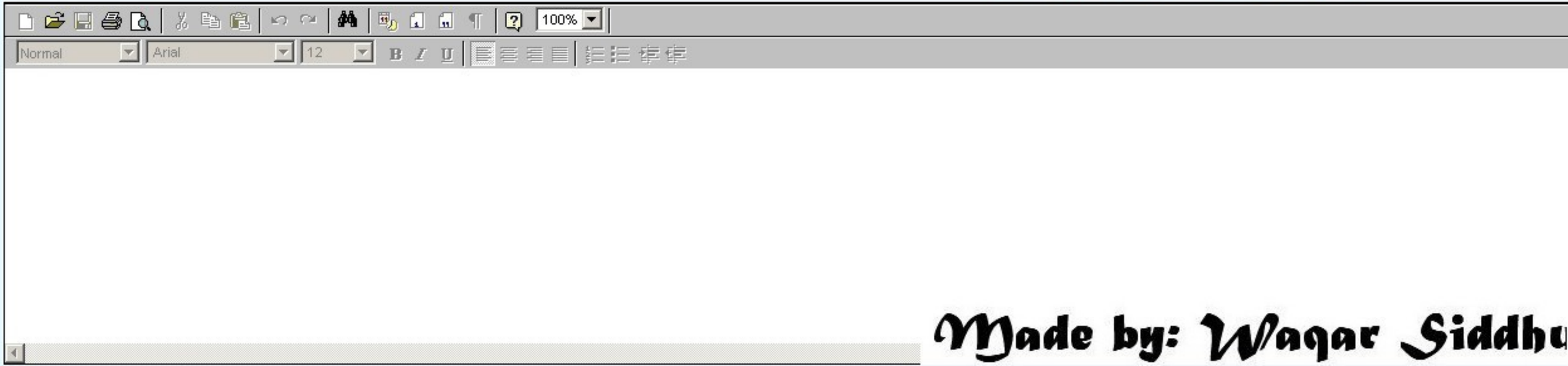
Question No : 50 of 52 Marks: 5 (Budgeted Time 10 Min)

Determine whether the following vector field $\vec{F}$ is conservative or not.

$\vec{F}(x, y, z) = x^2 z\,\hat{i} + y^2 x\,\hat{j} + (y + 2z)\,\hat{k}$

Answer (Please click here to Add Answer)

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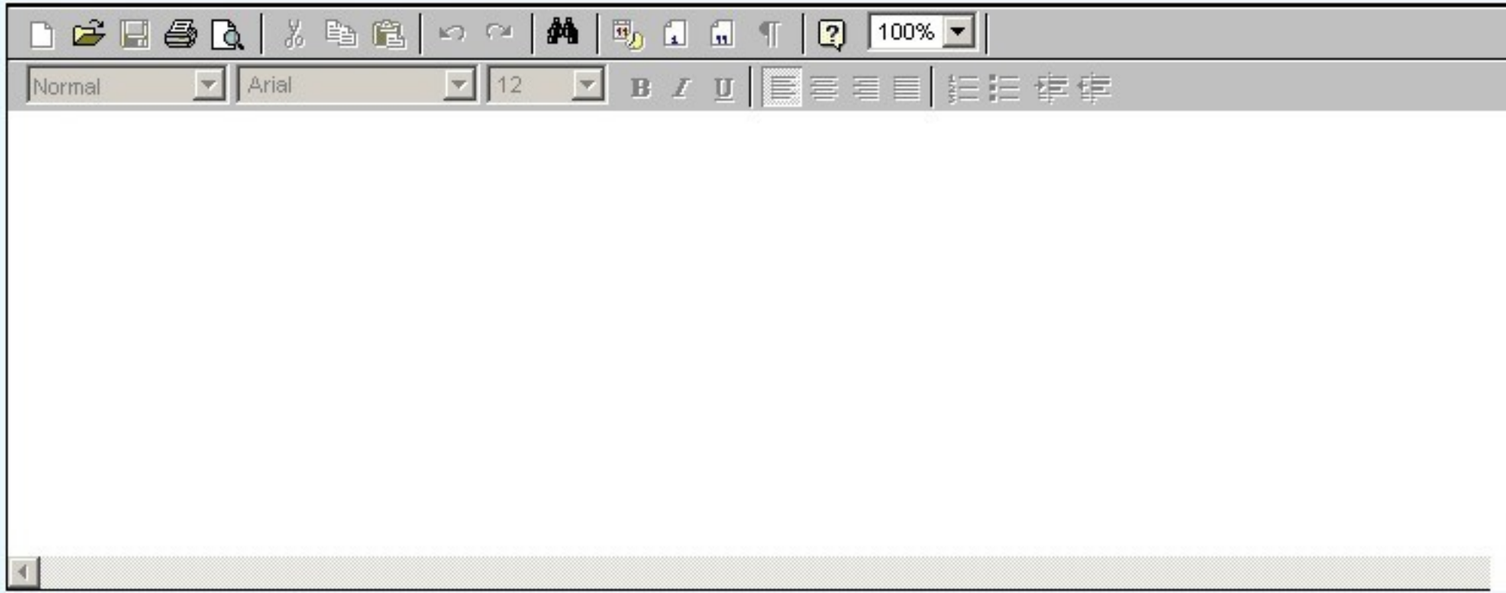
Question No : 51 of 52 Marks: 5 (Budgeted Time 10 Min)

If $f(x, y) = (x - 4)\ \ln(xy)$

Find both first order partial derivatives.

Answer (Please click here to Add Answer)

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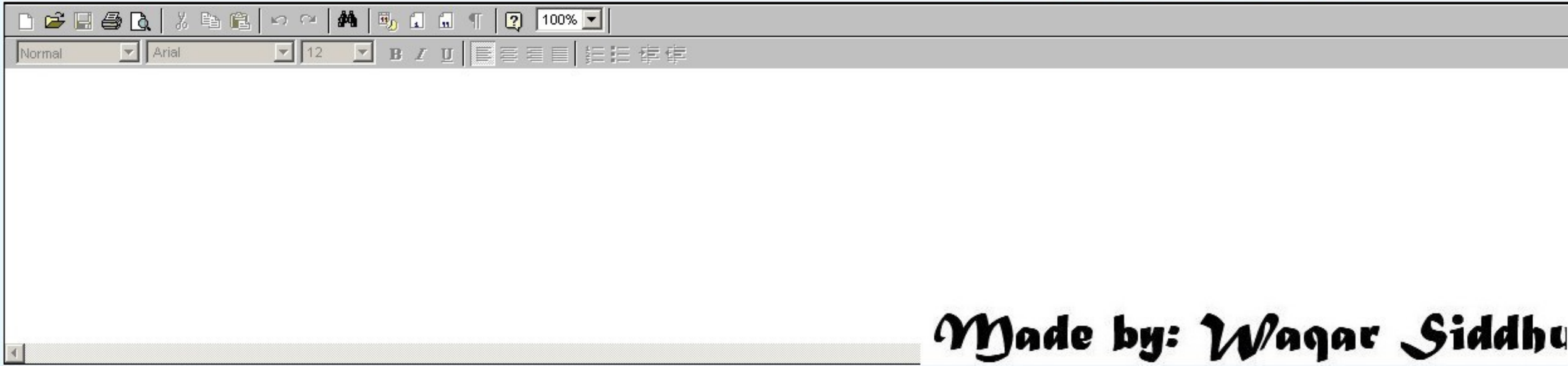
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Question No : 52 of 52 Marks: 5 (Budgeted Time 10 Min)

Let $\vec{r}(t) = t^2\,\hat{i} + t\,\hat{j} + (t^2 - 5)\,\hat{k}$. Find t, such that $\vec{r}(t)$ and $\vec{r'}(t)$ are perpendicular to each other.

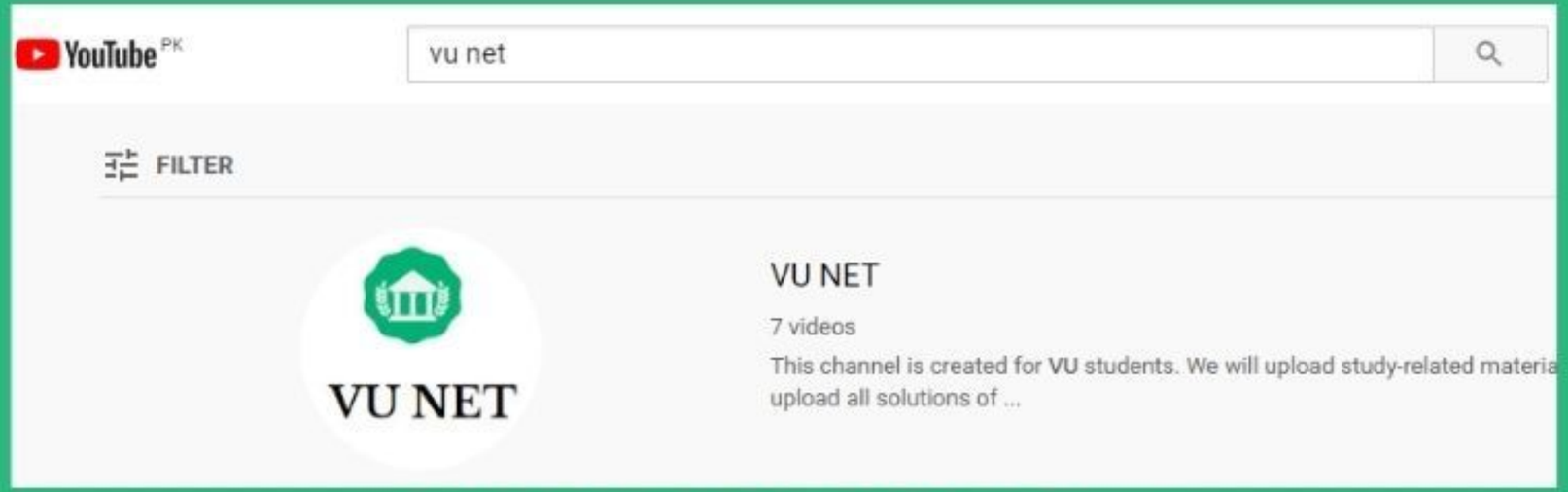
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